Which genetic test, when? A published case: infantile seizures solved by exome

Summary for doctors and healthcare teams, written in our own words from a published, peer-reviewed case report. Please read the original article (link below).

Source: Kadian J, Yadav D, Alhan P, Bhagwan J, Yadav D, Bhalla K. *Exploring role of genetics in diagnosing, management, and outcome of infantile epilepsy: A case series of three interesting cases.* J Family Med Prim Care. 2025;14(11):4891–4893. Department of Pediatrics, PGIMS Rohtak, India. Case 2 summarised.
pmc.ncbi.nlm.nih.gov/articles/PMC12705023

The patient

- 6-month-old boy
- Development normal until presentation
- Non-consanguineous parents; unremarkable birth history
- Seizures: increased tone of all limbs with eye up-rolling

Work-up before genetics

- MRI brain: normal
- EEG: normal
- Metabolic screen: negative
- Phenytoin: seizure-free for 3 months, then recurrence with the same semiology
- Levetiracetam titrated; clobazam added

The genetic test and result

Whole exome sequencing was ordered to look for a channelopathy. It found a frameshift variant in exon 2 of **PRRT2**, consistent with **self-limited (familial) infantile epilepsy (SeLIE)**, a condition the authors note is often confused with developmental and epileptic encephalopathies. The child became seizure-free on treatment.

Why the diagnosis matters

In PRRT2-related SeLIE, onset is usually at 4–7 months, seizures typically resolve by about 2 years of age, development is normal, and response to anti-seizure medication is usually excellent; inheritance is autosomal dominant with reduced penetrance (GeneReviews). A precise diagnosis therefore supports **reassurance, an accurate prognosis, family counselling and fewer unnecessary investigations**.

Talking points for the doctor visit

- Normal MRI/EEG/metabolic work-up does not rule out a genetic epilepsy.
- Exome (or an epilepsy panel of 25+ genes) is a first-tier option in unexplained epilepsy; exome/genome is often preferred because it more often finds the cause.
- A genetic diagnosis can be good news: it can confirm a self-limited condition with a good outlook.
- Send full clinical details; trio (child + parents) when possible; offer genetic counselling before and after.

For healthcare professionals' general awareness. Not a substitute for clinical judgement; facts and guidelines may change. Case details are as published by the original authors; consult the full article. Lab-specific details (sample, turnaround, report) as per your laboratory.

EPISODE 1 · QUICK GUIDE

Which genetic test, when?

Test	What it finds	Typical use / note
Karyotype	Chromosome number; balanced translocations; large rearrangements. Often misses imbalances of 5–10 Mb.	Obvious chromosomal syndrome (e.g. Down syndrome); recurrent miscarriage (parental balanced translocation); family history of rearrangement.
Chromosomal microarray (CMA)	Copy number variants (deletions/duplications) genome-wide. Misses balanced changes and low-level mosaicism.	First-tier for unexplained DD/ID, autism, multiple congenital anomalies (2010 ISCA consensus). Yield ~15–20% vs ~3% for karyotype.
Gene panel	Selected genes for one disease group.	When features clearly point to one group; in epilepsy, a panel of 25+ genes is a first-tier option.
Whole exome sequencing	Coding regions of ~20,000 genes (single-letter changes, small insertions/deletions).	~36% yield in neurodevelopmental disorders (meta-analysis); first-tier in many guidelines. Misses repeat expansions, mitochondrial DNA, many large CNVs.
Special tests	Fragile X: PCR for repeat expansion · SMA and other deletions: MLPA · Imprinting disorders: methylation testing.	Choose when the clinical question points to them.

Trio testing

Testing the patient with both parents makes de novo variants easy to identify and gives clearer reports.

Reading the report

Five categories (ACMG/AMP): pathogenic, likely pathogenic, uncertain significance (VUS), likely benign, benign. Do not base treatment on a VUS; classifications can change over time.

References

- Kadian J et al. J Family Med Prim Care 2025;14(11):4891–3. PMC12705023
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- Epilepsy genetic testing: Sheidley et al. (Epilepsia) review and Smith et al. practice guideline, summarised by Boston Children's Hospital. answers.childrenshospital.org/epilepsy-genetic-testing
- GeneReviews: PRRT2-associated paroxysmal disorders. NBK475803
- Richards S et al. Standards and guidelines for the interpretation of sequence variants (ACMG/AMP). Genet Med 2015.

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